Discharges w low D_2 MGI

gap 23.3.16

32006: 0.9 bar D_2 + Ar traces in 0.8 MA circular plasma, E_{th} = 0.13 MJ, q_{95} = 3.7, 2 MW ECRH

DCN H-1:
t = 1 s: n_e dl = 3.25 10^{19} / m^2 l_{H-1} ~ 1.10 m Vol ~ 10.5 m^3
t = 1.07-1.015 s: n_e dl = 5-5.5 10^{19} / m^2 l_{H-1} ~ 1.10 m Vol ~ 10.5 m^3 (**)
ΔN_e ~ 2 10^{19} / m^2 / (1.10 m) x 10.5 m^3 = 1.91 10^{20} n. electrons
N_{D2} = 0.9 bar x 0.08 l x 2.4 10^{22} = 10^{21} molecules
N_D = 2 x N_{D2} = 2. 10^{21} D atoms
assimilation fraction = ΔN_e / N_D ~ 10 % (before TQ)

From the profile reconstruction: n_e ~ 4.5 – 5 10^{19} / m^2 (**)

COO V-1:
t = 1 s: n_e dl = 3.2 10^{19} / m^2 l_{V-1} ~ 1.115 m Vol ~ 10.5 m^3
t = 1.07-1.015 s: n_e dl = 6.8 10^{19} / m^2 l_{V-1} ~ 1.115 m Vol ~ 10.5 m^3
ΔN_e ~ 3.4 10^{19} / m^2 / (1.115 m) x 10.5 m^3 = 3.20 10^{20} n. electrons
N_{D2} = 0.9 bar x 0.08 l x 2.4 10^{22} = 10^{21} molecules
N_D = 2 x N_{D2} = 2. 10^{21} D atoms
assimilation fraction = ΔN_e / N_D ~ 16 % (before TQ)

31025: 5 bar D_2 in 1 MA elongated plasma, ~ ohmic E_{th} = 0.18 MJ, q_{95} = 4.5, 0.7 MW ECRH

DCN H-1:
t = 2 s: n_e dl = 5.2 10^{19} / m^2 l_{H-1} ~ 1.05 m Vol ~ 13.5 m^3
not correct in CQ

COO V-1:
t = 2 s: n_e dl = 6. 10^{19} / m^2 l_{V-1} ~ 1.42 m Vol ~ 13.5 m^3
t = 2.004 s: n_e dl = 6. 10^{20} / m^2 l_{V-1} ~ 1.42 m Vol ~ 13.5 m^3
ΔN_e ~ 5.4 10^{20} / m^2 / (1.42 m) x 13.5 m^3 = 5.1 10^{21} n. electrons
N_{D2} = 5 bar x 0.08 l x 2.4 10^{22} = 0.96 10^{22} molecules
N_D = 2 x N_{D2} = 1.92 10^{22} D atoms
assimilation fraction = ΔN_e / N_D ~ 27 % (t = 2.004 s, not all gas injected ...)
assimilation fraction corrected = ΔN_e / (N_D 0.7) ~ 38 % (t = 2.004 s, only gas injected)

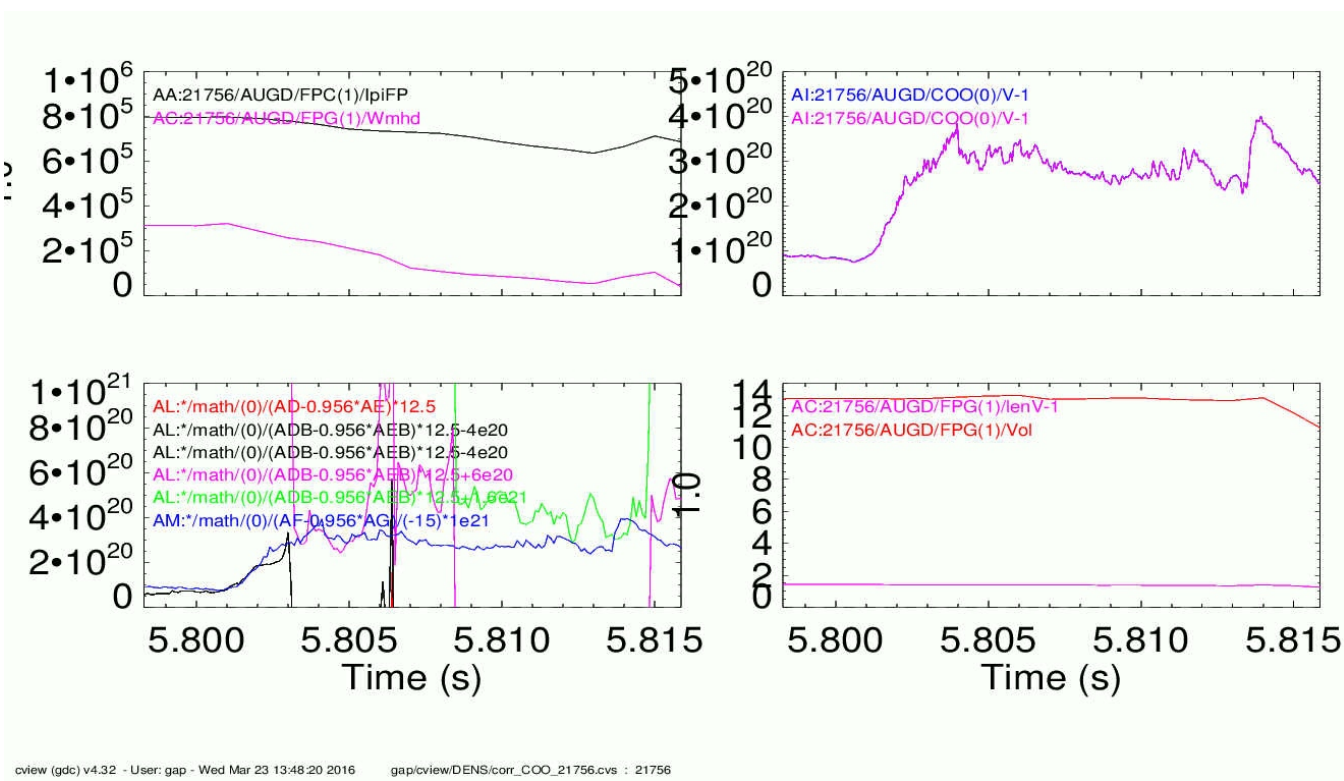
(consistent w density measured during CQ)

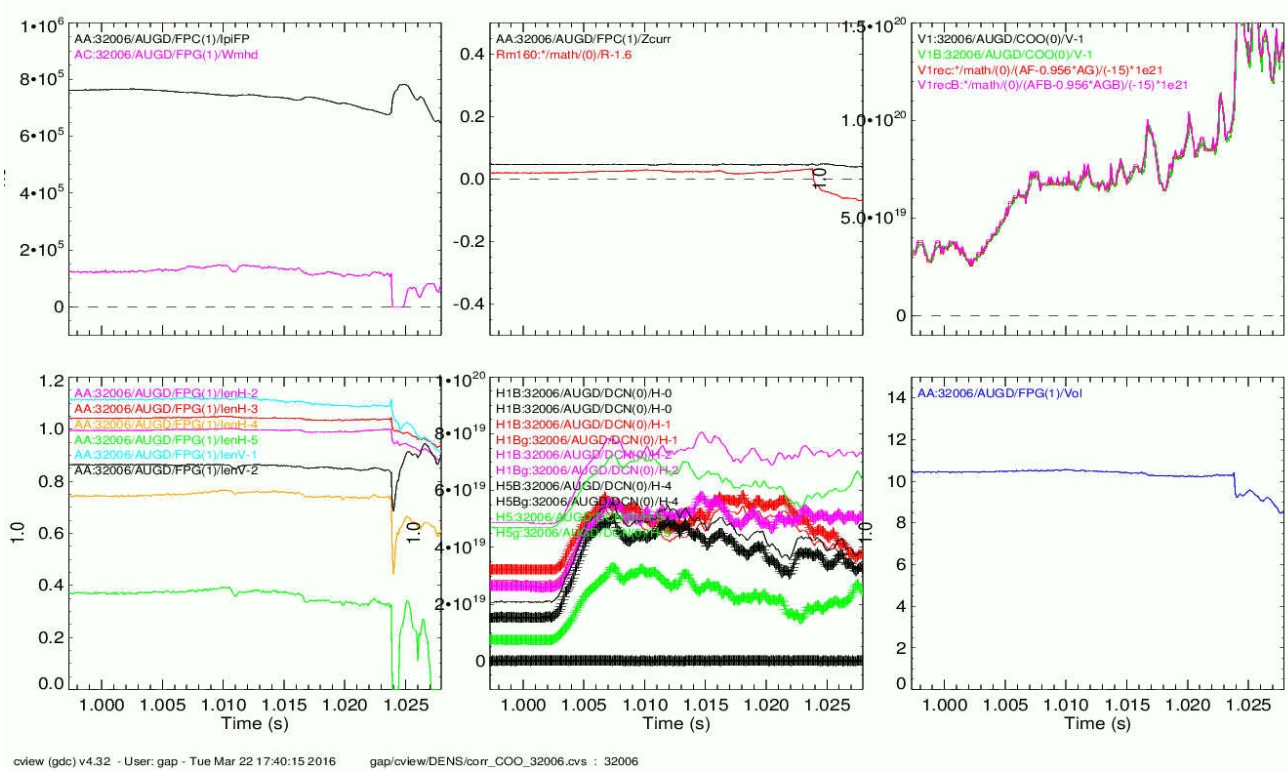
31026: 10 bar D2 in 1 MA elongated plasma (repeat of 31025)
consistent w 31025 (bad COO V-1)

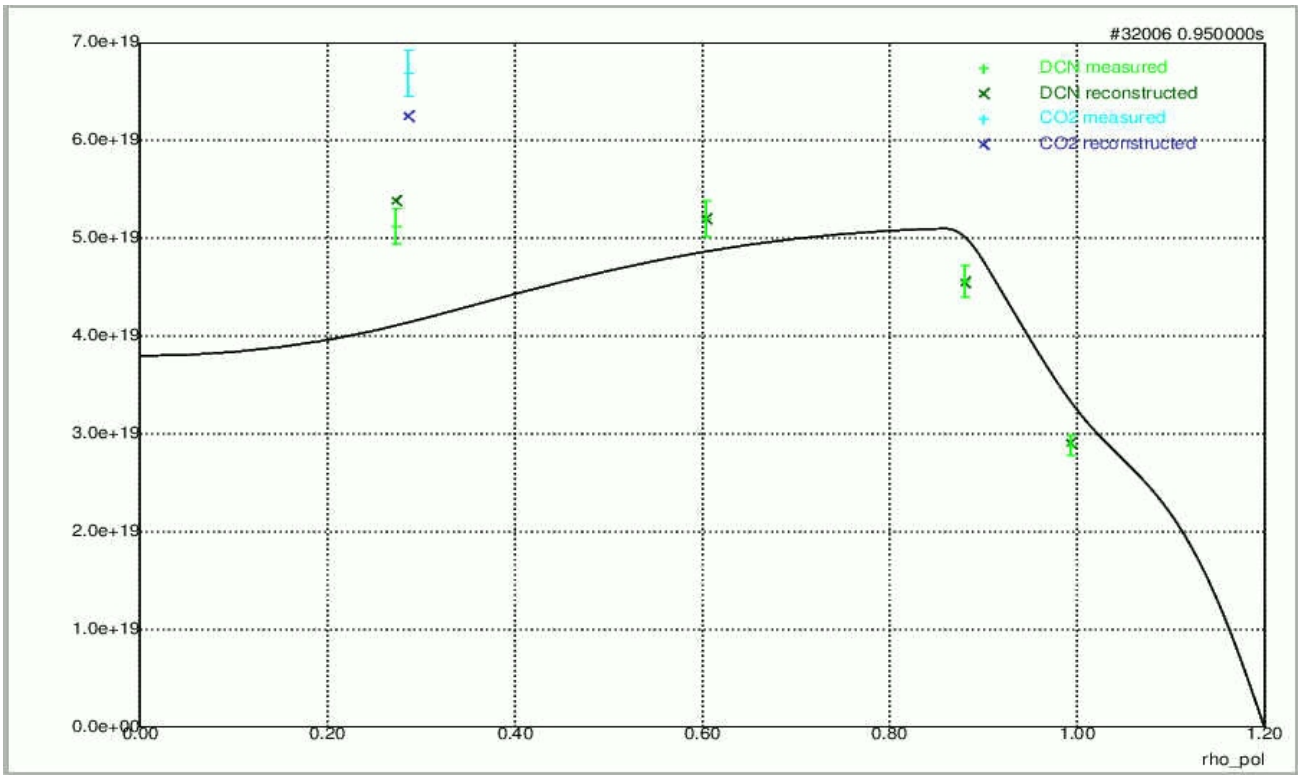
21756: 10 bar D_2 (40 m^3) in 0.8 MA elongated plasma
E_{th} = 0.33 MJ, q_{95} = 5.3, 5 MW NBI

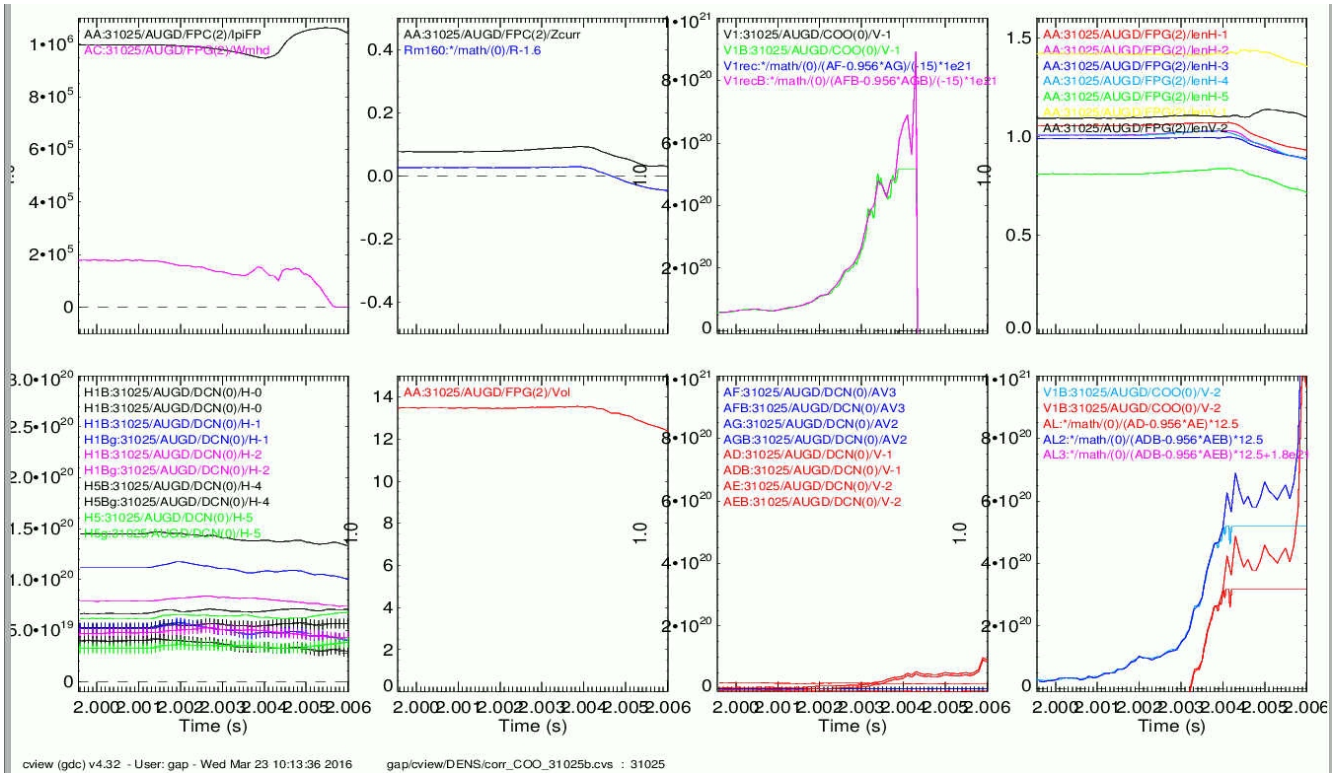
DCN H-1:
not correct in CQ

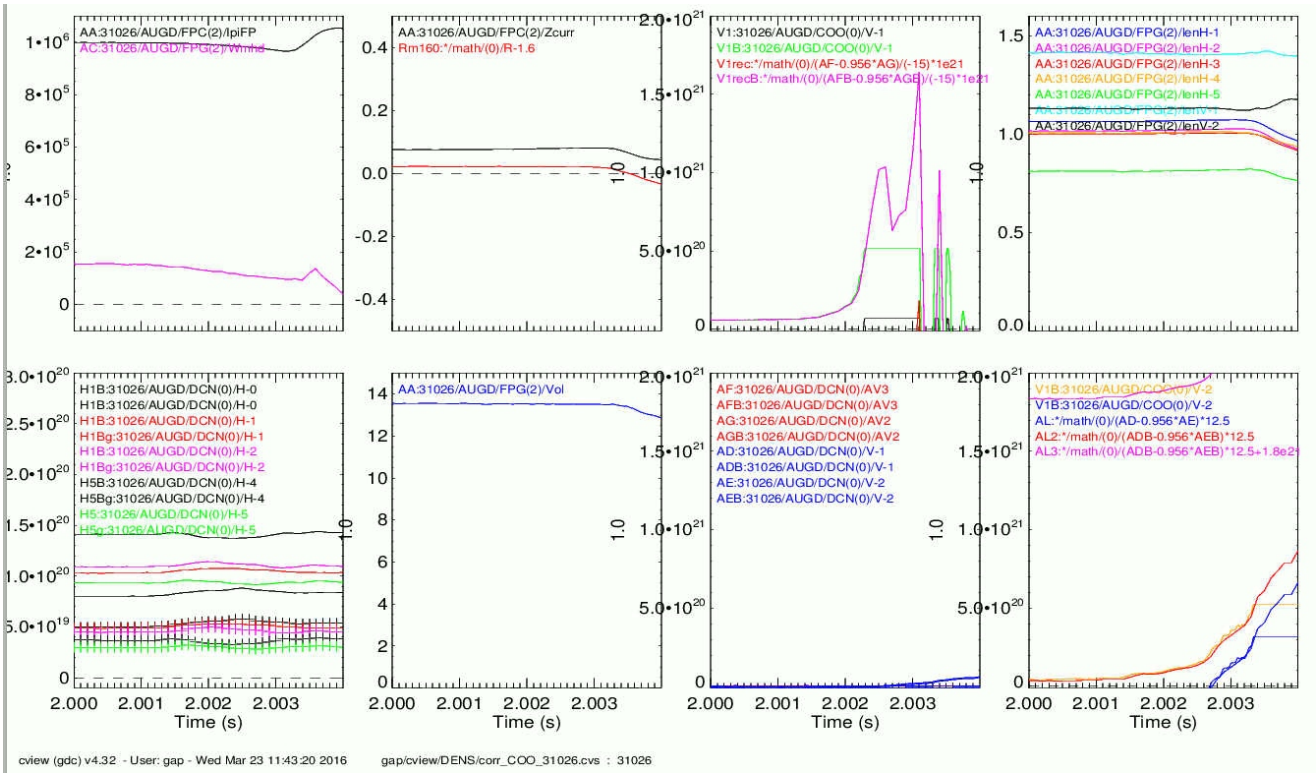
COO V-1:
t = 5.8 s: n_e dl = 8.5 10^{19} / m^2 l_{V-1} ~ 1.4 m Vol ~ 13 m^3
t = 5.805-5.81s: n_e dl = 3. 10^{20} / m^2 l_{V-1} ~ 1.4 m Vol ~ 13 m^3
ΔN_e ~ 2.15 10^{20} / m^2 / (1.4 m) x 13 m^3 = 2 10^{21} n. electrons
N_{D2} = 10 bar x 0.04 l x 2.4 10^{22} = 0.96 10^{22} molecules
N_D = 2 x N_{D2} = 1.92 10^{22} D atoms
assimilation fraction = ΔN_e / N_D ~ 10.5 %

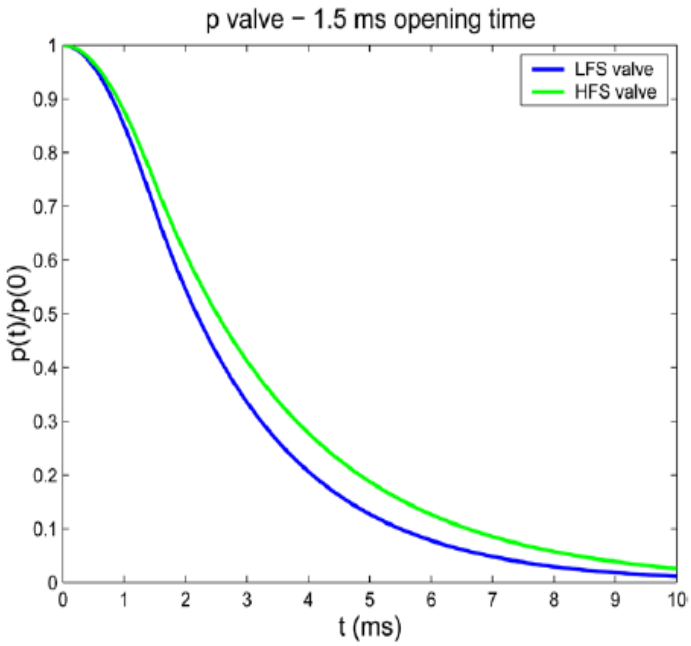


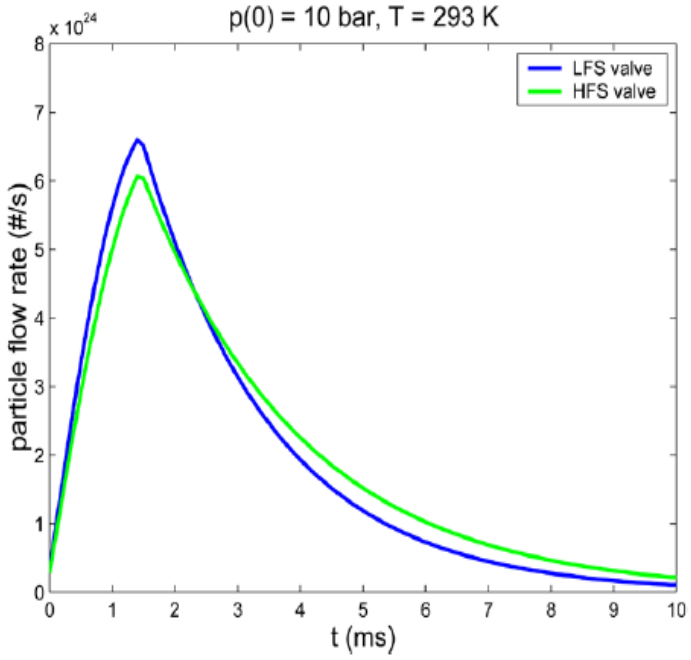

AA:32006/AUGD/FPC(1)/IpiFP
AC:32006/AUGD/FPG(1)/Wmhd
AA:32006/AUGD/FPC(1)/Zcurr
Rm160:*/math/(0)/R-1.6
V1:32006/AUGD/COO(0)/V-1
V1B:32006/AUGD/COO(0)/V-1
V1rec:*/math/(0)/(AF-0.956*AG)/(-15)*1e21
V1recB:*/math/(0)/(AFB-0.956*AGB)/(-15)*1e21
AA:32006/AUGD/FPG(1)/lenH-2
AA:32006/AUGD/FPG(1)/lenH-3
AA:32006/AUGD/FPG(1)/lenH-4
AA:32006/AUGD/FPG(1)/lenH-5
AA:32006/AUGD/FPG(1)/lenV-1
AA:32006/AUGD/FPG(1)/lenV-2
H1B:32006/AUGD/DCN(0)/H-0
H1B:32006/AUGD/DCN(0)/H-0
H1B:32006/AUGD/DCN(0)/H-1
H1Bg:32006/AUGD/DCN(0)/H-1
H1B:32006/AUGD/DCN(0)/H-2
H1Bg:32006/AUGD/DCN(0)/H-2
H5B:32006/AUGD/DCN(0)/H-4
H5Bg:32006/AUGD/DCN(0)/H-4
AA:32006/AUGD/FPG(1)/Vol
Time (s)
Time (s)
Time (s)
cview (gdc) v4.32 - User: gap - Tue Mar 22 17:40:15 2016
gap/cview/DENS/corr_COO_32006.cvs : 32006


#32006 0.950000s
DCN measured
DCN reconstructed
CO2 measured
CO2 reconstructed
rho_pol


AA:31025/AUGD/FPC(2)/IpiFP
AC:31025/AUGD/FPG(2)/Wmhd
AA:31025/AUGD/FPC(2)/Zcurr
Rm160:*/math/(0)/R-1.6
V1:31025/AUGD/COO(0)/V-1
V1B:31025/AUGD/COO(0)/V-1
V1rec:*/math/(0)/(AF-0.956*AG)/(-15)*1e21
V1recB:*/math/(0)/(AFB-0.956*AGB)/(-15)*1e21
AA:31025/AUGD/FPG(2)/lenH-1
AA:31025/AUGD/FPG(2)/lenH-2
AA:31025/AUGD/FPG(2)/lenH-3
AA:31025/AUGD/FPG(2)/lenH-4
AA:31025/AUGD/FPG(2)/lenH-5
AA:31025/AUGD/FPG(2)/lenV-2
H1B:31025/AUGD/DCN(0)/H-0
H1B:31025/AUGD/DCN(0)/H-0
H1B:31025/AUGD/DCN(0)/H-1
H1Bg:31025/AUGD/DCN(0)/H-1
H1B:31025/AUGD/DCN(0)/H-2
H1Bg:31025/AUGD/DCN(0)/H-2
H5B:31025/AUGD/DCN(0)/H-4
H5Bg:31025/AUGD/DCN(0)/H-4
H5:31025/AUGD/DCN(0)/H-5
H5g:31025/AUGD/DCN(0)/H-5
AA:31025/AUGD/FPG(2)/Vol
AF:31025/AUGD/DCN(0)/AV3
AFB:31025/AUGD/DCN(0)/AV3
AG:31025/AUGD/DCN(0)/AV2
AGB:31025/AUGD/DCN(0)/AV2
AD:31025/AUGD/DCN(0)/V-1
ADB:31025/AUGD/DCN(0)/V-1
AE:31025/AUGD/DCN(0)/V-2
AEB:31025/AUGD/DCN(0)/V-2
V1B:31025/AUGD/COO(0)/V-2
V1B:31025/AUGD/COO(0)/V-2
AL:*/math/(0)/(AD-0.956*AE)*12.5
AL2:*/math/(0)/(ADB-0.956*AEB)*12.5
AL3:*/math/(0)/(ADB-0.956*AEB)*12.5+1.8e2
Time (s)
cview (gdc) v4.32 - User: gap - Wed Mar 23 10:13:36 2016
gap/cview/DENS/corr_COO_31025b.cvs : 31025


AA:31026/AUGD/FPC(2)/IpiFP
AC:31026/AUGD/FPG(2)/Wmhd
AA:31026/AUGD/FPC(2)/Zcurr
Rm160:*/math/(0)/R-1.6
V1:31026/AUGD/COO(0)/V-1
V1B:31026/AUGD/COO(0)/V-1
V1rec:*/math/(0)/(AF-0.956*AG)/(-15)*1e21
V1recB:*/math/(0)/(AFB-0.956*AGB)/(-15)*1e21
AA:31026/AUGD/FPG(2)/lenH-1
AA:31026/AUGD/FPG(2)/lenH-2
AA:31026/AUGD/FPG(2)/lenH-3
AA:31026/AUGD/FPG(2)/lenH-4
AA:31026/AUGD/FPG(2)/lenH-5
AA:31026/AUGD/FPG(2)/lenV-2
H1B:31026/AUGD/DCN(0)/H-0
H1B:31026/AUGD/DCN(0)/H-0
H1B:31026/AUGD/DCN(0)/H-1
H1Bg:31026/AUGD/DCN(0)/H-1
H1B:31026/AUGD/DCN(0)/H-2
H1Bg:31026/AUGD/DCN(0)/H-2
H5B:31026/AUGD/DCN(0)/H-4
H5Bg:31026/AUGD/DCN(0)/H-4
H5:31026/AUGD/DCN(0)/H-5
H5g:31026/AUGD/DCN(0)/H-5
AA:31026/AUGD/FPG(2)/Vol
AF:31026/AUGD/DCN(0)/AV3
AFB:31026/AUGD/DCN(0)/AV3
AG:31026/AUGD/DCN(0)/AV2
AGB:31026/AUGD/DCN(0)/AV2
AD:31026/AUGD/DCN(0)/V-1
ADB:31026/AUGD/DCN(0)/V-1
AE:31026/AUGD/DCN(0)/V-2
AEB:31026/AUGD/DCN(0)/V-2
V1B:31026/AUGD/COO(0)/V-2
V1B:31026/AUGD/COO(0)/V-2
AL:*/math/(0)/(AD-0.956*AE)*12.5
AL2:*/math/(0)/(ADB-0.956*AEB)*12.5
AL3:*/math/(0)/(ADB-0.956*AEB)*12.5+1.8e2
Time (s)
cview (gdc) v4.32 - User: gap - Wed Mar 23 11:43:20 2016
gap/cview/DENS/corr_COO_31026.cvs : 31026


p valve − 1.5 ms opening time
LFS valve
HFS valve
p(t)/p(0)
t (ms)


p(0) = 10 bar, T = 293 K
x 10^24
LFS valve
HFS valve
particle flow rate (#/s)
t (ms)

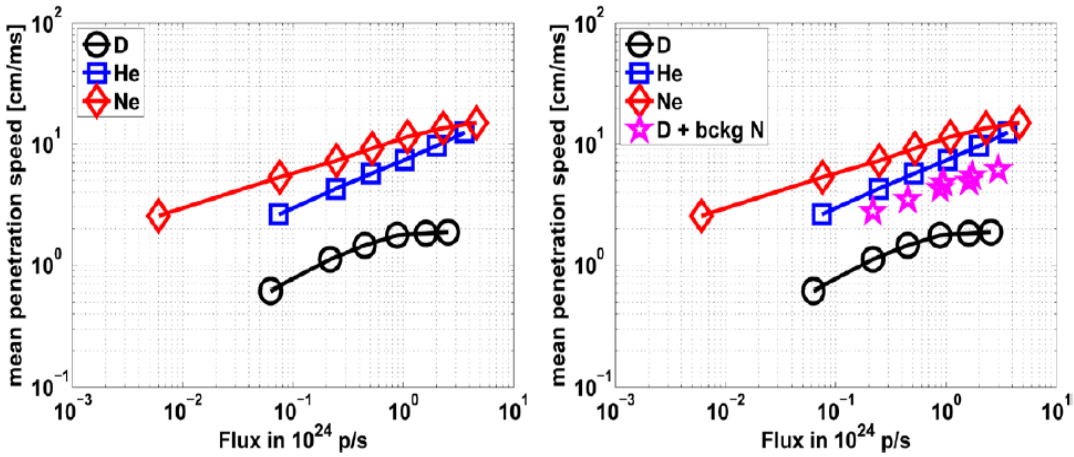


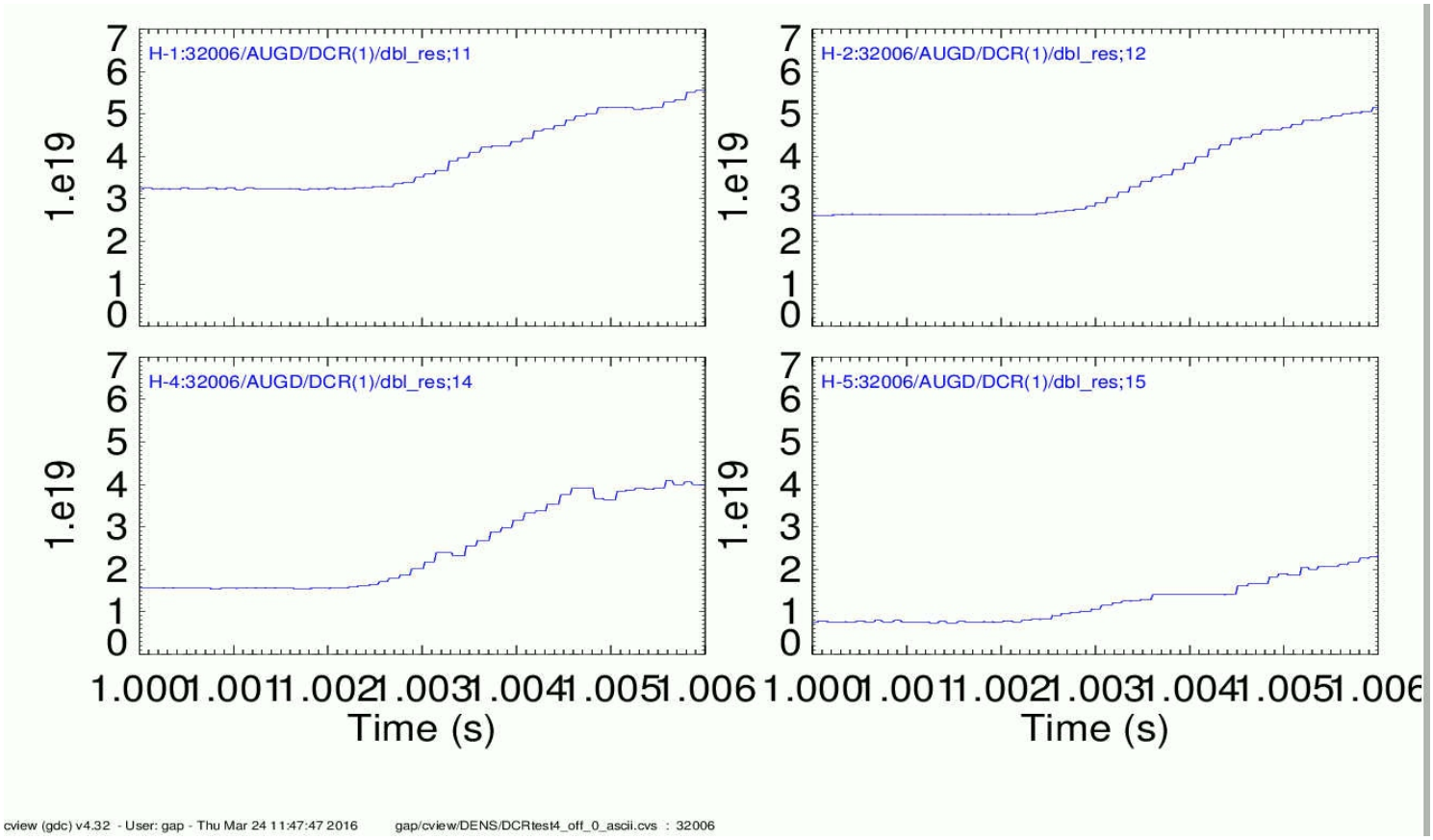
Figure 25. Left plot: comparison of cold front penetration speed (averaged over 1 ms time) versus the gas–rate. Different gas species are indicated in the legend. Right plot: same as left plot, but with addition of D MGI on top of background presence of N.

Remark (1) Please: reflect on this figure!

Remark (2) : what is the role of the fraction of injected gas, not assimilated, in the energy balance?

Remark (3) :
check vessel pressure to be sure of the amount of gas injected
check DCN data w Alexander's method - done
DCN geometry, time traces of line integrated densities – done

Density data prepared for shot 32006 (attention: circular plasma, integration path shorter than in elongated plasma)
files:
/u/gap/JOREK_DIR/D2_DISCUSSION/FIGS/ne_32006.txt
(fig. next page)
/u/gap/JOREK_DIR/D2_DISCUSSION/density.coords



cview (gdc) v4.32 - User: gap - Thu Mar 24 11:47:47 2016 gap/cview/DENS/DCRtest4_off_0_ascii.cvs : 32006